



Suppression of boron diffusion using carbon co-implantation in DRAM



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ABSTRACT

In this paper, germanium pre-amorphization implantation (PAI) and carbon co-implantation (Ge + C co-IIP) were applied to suppress boron diffusion. The corresponding characteristics were investigated in terms of the dopant diffusion, device performance, and its application to dynamic random access memory (DRAM). A shallow dopant profile was indicated and the threshold voltage (V_{TH}) was reduced by approximately 45 mV by Ge + C co-IIP. In the DRAM device, the V_{TH} mismatch of the sense amplifier NMOS pairs was reduced by approximately 15% and the write characteristics were improved two-fold.

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1. Introduction

As the minimum feature size of silicon (Si) complementary metal-oxide-semiconductor (CMOS) devices shrinks to the nanometer range, it is a challenge to suppress the short channel effect (SCE) degrading transistor performance, due to new physical phenomena at short dimensions and fundamental limitations in the material properties. Although the threshold voltage (V_{TH}) roll-off, which is one of SCE problems, can be suppressed by locally increasing the channel doping concentration near the drain or source/drain (S/D) junctions [1] (e.g., halo [2] or pocket implantation [3]), the dopants diffuse randomly and deeply because of transient enhanced diffusion (TED), eventually causing severe variation in V_{TH} . To suppress the diffusion of halo dopants, several technologies have been investigated, such as cold implantation [4], carbon (C) or fluorine (F) co-implantation [5,6], and heavy dopant (indium to NMOS) implantation [7]. Since cold implantation is effective only above a critical dose where an amorphous layer can be formed [8] and the indium implantation increases the sheet resistance due to the acceptor freeze-out effect [9],

co-implantation technology seems to be optimal for halo implantation. In this paper, we investigate the effects of Ge + C co-IIP in terms of the dopant diffusion and device performance through secondary ion mass spectroscopy (SIMS), transmission electron microscopy (TEM), and electrical analyses. For DRAM applications, we also discuss the influence of Ge + C co-IIP on V_{TH} mismatch of the sense amplifier NMOS pairs and on failbits as a function of write delay time.

2. Experiments

For the fabrication of an N-channel metal oxide semiconductor field effect transistor (NMOSFET), boron (B) halo implantation was performed after the gate was etched and a lightly doped drain (LDD) region was formed. The sidewall spacer was formed and S/D N+ (As) implantation was performed. To examine the effects of C that suppresses B diffusion, SIMS profiles were analyzed before and after 1050 °C 10 s spike rapid thermal annealing (RTA) for samples performing only Ge co-IIP and Ge + C co-IIP prior to B halo implantation. To assess the effects of Ge PAI, the formation of an amorphous layer was compared using TEM analysis between the C co-IIP sample and Ge + C co-IIP. To understand the effects of Ge + C co-IIP on the DRAM device, halo implantation was performed for one wafer and Ge + C co-IIP before the halo process for another wafer while the other processes were kept the same. V_{TH} and on/off

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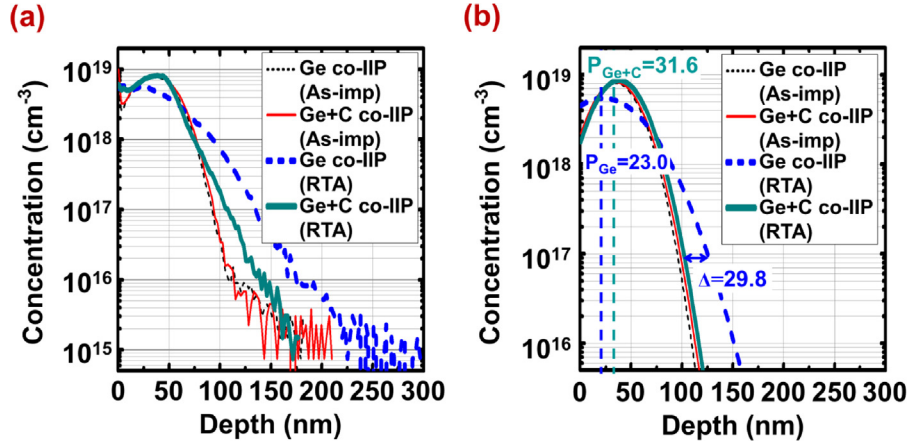


Fig. 1. (a) SIMS profiles comparison of B atoms between Ge co-IIP and Ge+C co-IIP for the as-implanted and after RTA case. (b) Calculated SIMS profiles.

currents were measured for the whole wafer, and failbits with reduced write delay time were measured.

3. Results and discussion

According to the SIMS analysis in Fig. 1, in the as-implanted profiles, the calculated peak position were 34.5 and 35.1 nm and the standard deviations were 20.4 and 21.2 nm for Ge co-IIP and Ge+C co-IIP, respectively. There was no change via the addition of C atoms. However, in these profiles after RTA, the calculated peak position and diffusivity (D) were 23.0 nm (P_{Ge}) and 4.45×10^{-13} cm²/s for the Ge co-IIP case and 31.6 nm (P_{Ge+C}) and 2.54×10^{-22} cm²/s for the Ge+C co-IIP case. Because the peak position shifted left and the diffusion distance (Δ) increased by approximately 29.8 nm for the Ge co-IIP case, it was confirmed that the diffusion of B dopants was suppressed and a more shallow profile was formed by Ge+C co-IIP. Graoui et al. [10] reported shallower profiles for Ge+C co-IIP than those of Ge co-IIP. Co-implanted C atoms can form clusters with Si interstitials, such as IC (mobile) and IC2 (immobile), and the TED of the B atoms was reduced and the B diffusion was suppressed accordingly. [5,11] Based on a study of Pawlak et al. [12], C atoms have to be located at the substitutional site by Ge PAI, which is necessary to form clusters between Si interstitials and C atoms. In Fig. 2, the formation of the amorphous layer due to Ge PAI was confirmed by TEM analysis in the Ge+C co-IIP sample before heat treatment.

Therefore, the Ge+C co-IIP condition was suitable for suppressing B diffusion.

In Fig. 3, the device performance was investigated due to Ge+C co-IIP. Based on the correlation between V_{TH} and the off current (I_{off}), the V_{TH} of the Ge+C co-IIP sample was less than that of the control sample at the same I_{off} value. According to the I_{off} equation in (1), low V_{TH} at the same I_{off} means the increment in the effective channel length (L_{eff}) if all other parameters are the same. [13]

$$I_{off} [nA] = 100 \times \frac{W}{L_{eff}} \times e^{\frac{qV_{TH}}{\eta kT}}, \eta = 1 + \frac{C_{dep}}{C_{ox}} \quad (1)$$

Additionally, in Fig. 3(b), the off current was slightly increased as a result of the V_{TH} reduction, but the on current remained the same since the V_{TH} reduction was canceled by the L_{eff} increment. Therefore, the L_{eff} was increased and the drain induced barrier lowering (DIBL) was relatively improved through the suppression of B diffusion by Ge+C co-IIP. In addition, as shown in Fig. 3(a), the measurements over the whole wafer were narrowly distributed in the Ge+C co-IIP case and the degree of scattering was reduced by approximately 18 mV, from 57 (control) to 39 mV (Ge+C co-IIP). Therefore, V_{TH} variation by B diffusion was effectively suppressed through the Ge+C co-IIP.

When the effects were applied to the DRAM device, in which L_{eff} increased and the V_{TH} variation was reduced due to Ge+C co-IIP, the V_{TH} mismatch of the pair transistors in the sense amplifier

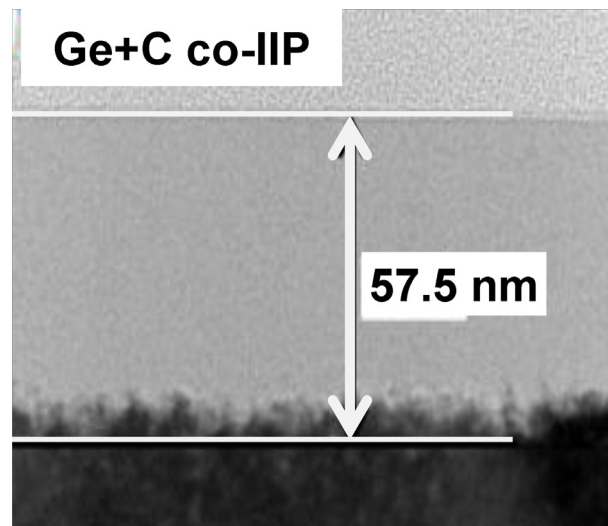


Fig. 2. Cross-sectional TEM image of the Ge+C co-implanted sample before annealing (bright gray area indicates an amorphized region with a thickness of ~57.5 nm).

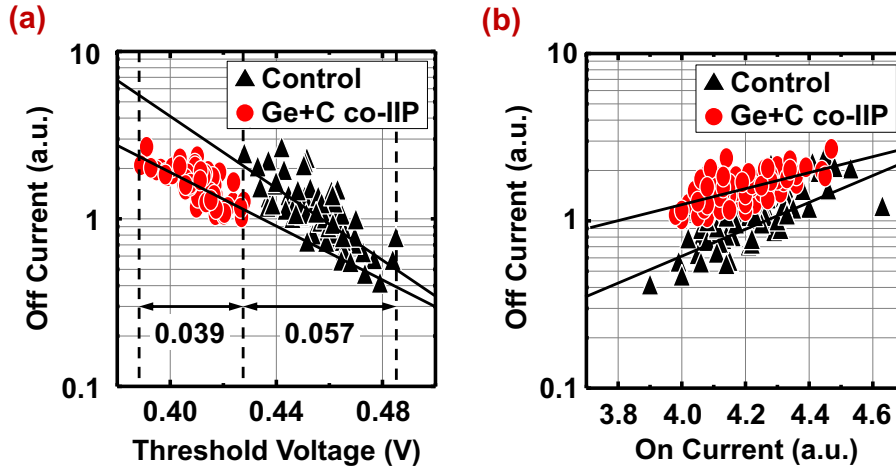


Fig. 3. (a) V_{TH} vs. the off current correlation and (b) on/off current correlation with and without Ge + C co-IIP.

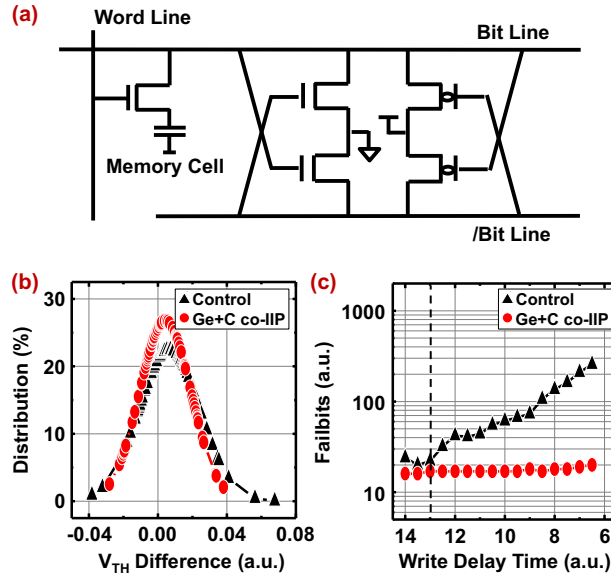


Fig. 4. (a) Schematics of the sense amplifier circuit. (b) The distribution of the threshold voltage mismatch of the NMOS pairs in the sense amplifier with and without Ge + C co-IIP. (c) Failbits with reduced write delay time.

circuit decreased and the performance of the sense amplifier improved. V_{TH} mismatch was defined as the difference in V_{TH} between the pair transistors. In the DRAM device, the charge sharing between the cell capacitance and bit line capacitance produced a slight voltage difference (less than 0.2 V) and the sense amplifier circuit, as shown in Fig. 4(a), detected and amplified this voltage difference. [14] Since V_{TH} mismatch causes an imbalance in this latch-type sense amplifier that can amplify the voltage difference in the wrong direction, V_{TH} mismatch should be minimized to improve the performance of the sense amplifier. [15] However, V_{TH} mismatch is increased by factors such as the variation of the channel area, diffusion of the halo dopants and random dopant fluctuation (RDF) of the channel dose. [16] As shown in the V_{TH} mismatch distribution for the NMOS pairs in the sense amplifier (Fig. 4(b)), the distribution was narrowed by approximately 15% (53 → 45 mV, according to the 3-sigma value) and the peak point was closed to “0” (0.007 → 0.004) for the Ge + C co-IIP case. From these results, V_{TH} mismatch of the NMOS pairs was reduced through Ge + C co-IIP. If the write delay time was reduced, “write fail” occurred because data could be sufficiently

written to the memory cell capacitor. However, if the V_{TH} mismatch was reduced, the probability that this insufficient data could be amplified correctly was increased and the likelihood of “write fail” was reduced. In the Ge + C co-IIP case, the failbits did not increase in spite of shortening the write delay time and the margin for the write delay time was doubled, as shown in Fig. 4(c). Therefore, the performance of the sense amplifier and write characteristics was effectively improved by Ge + C co-IIP.

4. Conclusion

After the Ge + C co-IIP was applied, the suppression of B diffusion was investigated by SIMS, TEM and electrical measurements. TED was suppressed, accordingly reducing the B diffusion that was used for halo implantation. With this effect, the effective channel length was increased and the DIBL was improved. V_{TH} variation was also reduced for the NMOS transistor. In the DRAM application, V_{TH} mismatch of the sense amplifier NMOS pairs was reduced and the write characteristics were improved considerably by Ge + C co-IIP.

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